IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-093

1. Topic (12 words max)

Transforming Conventional Farming into an Intelligent Smart Farm

2. Research group the project belongs to

MR - Mixed Reality

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-093
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Traditional livestock farming systems still rely heavily on manual monitoring, conventional feeding methods, and experience-based decision-making. These practices often lead to inefficient resource utilization, delayed disease identification, inconsistent feeding management, reproductive losses, and poor market forecasting. As global demand for livestock products increases, farmers face growing pressure to improve productivity while maintaining animal welfare and reducing operational costs.

One of the major challenges in traditional farms is the inability to continuously monitor animal health and behavior in real time. Diseases are often identified only after visible symptoms appear, increasing mortality rates and treatment costs. In addition, feeding processes are commonly generalized rather than personalized, resulting in nutritional imbalances and feed wastage. Reproductive management is also difficult due to the lack of predictive systems capable of identifying healthy birth or hatching conditions in advance. Furthermore, farmers frequently struggle with fluctuating livestock market prices and lack accurate forecasting tools to support profitable selling and buying decisions.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and Computer Vision have shown significant potential in addressing these challenges. However, most existing solutions focus only on individual farming functions rather than providing a fully integrated smart livestock management platform.

Therefore, this research proposes an AI/ML-powered automated smart farm system capable of integrating intelligent feeding, disease detection, birth and hatch prediction, and AI-based market forecasting into a unified platform. The proposed system aims to improve farm efficiency, animal welfare, productivity, and financial sustainability through data-driven automation and predictive analytics.

6. Brief description of Existing Research and Systems (300 words max)

Existing research in smart agriculture and livestock farming has introduced several AI-based systems focusing on specific farm management tasks. Many studies have explored precision livestock farming using IoT sensors, wearable devices, and computer vision techniques to monitor animal health, feeding behavior, and environmental conditions. AI-driven disease detection systems have been developed to identify abnormalities such as reduced movement, abnormal posture, or changes in temperature through image processing and sensor analysis.

Several automated feeding systems currently use RFID tags and sensor technologies to provide scheduled feeding and monitor feed consumption. However, most of these systems do not provide personalized feeding recommendations based on animal health conditions, size, or behavior patterns. Similarly, research related to reproductive prediction mainly focuses on isolated prediction models for pregnancy monitoring or egg hatch analysis using environmental and historical data.

Market prediction systems also exist in agricultural sectors where machine learning algorithms are used to forecast crop or livestock prices using historical market trends. However, these systems generally operate independently and are not integrated with farm management platforms.

Although these solutions demonstrate the effectiveness of AI and IoT technologies in agriculture, current systems have several limitations. Most existing solutions are designed for a single functionality, lack real-time integration, and require manual intervention for decision-making. In addition, many systems do not support centralized monitoring or adaptive learning capabilities.

The proposed research differs from existing systems by introducing a unified AI-powered smart livestock platform that integrates multiple intelligent modules including personalized feeding, disease detection, birth and hatch prediction, and live market forecasting. By combining real-time sensor data, computer vision, predictive analytics, and machine learning models into one ecosystem, the system aims to provide a more scalable, efficient, and proactive livestock management solution.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an AI/ML-powered automated smart livestock management system designed to modernize traditional farming practices using intelligent technologies. The system integrates Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), Computer Vision, and Cloud-based analytics to automate critical livestock farming operations.

The solution consists of four main intelligent modules:

1. **Automated Feeding and Nutrition Management**
   Cameras, sensors, and animal identification techniques are used to monitor animal characteristics such as species, size, weight, and health condition. Based on collected data, machine learning algorithms generate personalized feeding recommendations. Automated feeding mechanisms then dispense the correct quantity of food while continuously optimizing feeding strategies.

2. **Disease Detection and Emergency Alert System**
   Cameras and IoT sensors continuously monitor animal movement, posture, temperature, and behavior patterns. AI models analyze these inputs to identify abnormal conditions or early disease symptoms. When abnormalities are detected, the system generates real-time alerts and provides initial recommendations to farmers or veterinarians.

3. **Birth and Egg Hatch Prediction System**
   Historical breeding records, environmental data, and imaging data are analyzed using predictive machine learning models. The system estimates the probability of healthy calf birth or successful egg hatching while identifying possible risks. Early warnings and preventive recommendations help improve reproductive success rates.

4. **AI-Based Market Prediction and Recommendation System**
   Real-time and historical market data related to livestock products are continuously collected and analyzed. Machine learning models forecast future market prices, demand fluctuations, and seasonal trends. Based on these predictions, the system recommends optimal buying or selling decisions to maximize profitability.

The system architecture follows a centralized smart farming model where sensor devices and cameras collect real-time farm data. The collected data is transmitted to a cloud-based platform where AI/ML models process and analyze information. The processed outputs are then visualized through dashboards and mobile applications, enabling farmers to monitor farm operations remotely and make data-driven decisions.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of the proposed AI-powered smart livestock farming system requires multidisciplinary expertise in livestock management, veterinary science, Artificial Intelligence (AI), Machine Learning (ML), and software engineering. The research team possesses the technical knowledge required to design and develop intelligent software systems, integrate sensor technologies, and implement AI-based predictive models.

Domain expertise related to animal health, nutrition, and reproductive management is essential for building accurate and practical system functionalities. Veterinary knowledge supports the identification of disease symptoms, abnormal animal behaviors, and appropriate preventive actions. Knowledge related to animal nutrition and feeding practices is also required to develop personalized feeding recommendation algorithms that improve animal growth and health conditions.

The project additionally requires expertise in data science and machine learning for training predictive models related to disease detection, birth and hatch prediction, and market forecasting. Computer vision techniques such as image classification and object detection are used to analyze animal posture, movement, and physical conditions. IoT expertise is necessary for integrating sensors that collect environmental and physiological data including temperature, humidity, activity levels, and feeding patterns.

The availability of data is a major strength of this research. The project plans to utilize datasets collected from National Livestock Development Board (NLDB) farms. These datasets include approximately 3450–3500 records related to livestock management, animal health, feeding patterns, reproductive data, and environmental conditions. Additional image datasets from Kaggle and other open-source repositories will be used to support AI model training for disease detection and animal monitoring.

Historical market data related to livestock prices, milk production, eggs, and meat sales will also be collected from online agricultural marketplaces and government agricultural databases. These datasets will support the development of AI-based market prediction models.

Modern development tools and technologies such as Python, TensorFlow, YOLO, OpenCV, MongoDB, Node.js, React, and cloud platforms will be used to implement the proposed system. The combination of technical expertise, domain knowledge, and accessible datasets provides a strong foundation for developing an intelligent, scalable, and practical smart farming solution capable of improving productivity, sustainability, and profitability in livestock farming.

9. Objectives and Novelty

Main Objective To design and develop an integrated AI/ML-powered smart farm system that automates core livestock farming operations—birth prediction, disease detection, nutrition management, and market interaction—enhancing animal welfare, productivity, and profitability.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22136138	Monitor animals via cameras, recommend and auto-feed food based on type, size, and health condition	• Real-Time Animal Monitoring • Animal Identification & Health Assessment • Smart Feeding Recommendation. • Automated Feeding & Data Optimization	Personalized Animal Diet Recommender using size estimation + weight + species classification + health logs via a recommendation system.
IT22282118	Use cameras and sensors to detect abnormal behavior or symptoms in animals. Recommend	• Monitor Animal Behavior in Real-Time • Detect Abnormal Symptoms Using AI	Real-time Behavior Anomaly Detection

	treatment or notify vet.	• Generate Alerts and Notifications. • Recommend Initial Treatment or Action.	
IT22139894	Predict whether a calf/egg will be born healthy or not using image, sensor, or historical data.	• Data Collection from Multiple Sources. • AI/ML-Based Analysis and Pattern Recognition. • Health Outcome Prediction. • Alerts and Preventive Action Suggestions	Multi-modal Predictive Birth Model: Combine sensor data (temperature, movement) and image data (ultrasound or egg images) to predict birth outcomes or egg hatchability
IT22134608	Maintain an up-to date market (selling/buying) with AI-based price prediction and recommendations.	• Collect and Update Market Data. • Analyze Market Trends Using AI/ML. • Predict Future Prices. • Provide Actionable Recommendations.	Customized real-time Market Demand Forecasting Model

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22136138	Design an AI-powered system that analyzes each animal's physical features, species, health status, and nutritional needs to recommend and automatically dispense personalized food and supplements. This ensures balanced feeding, reduces waste and optimizes animal growth and health.
IT22282118	Develop a system that automatically detects early signs of illness or distress in farm animals using cameras, audio sensors, and health history, and then provides recommendations or alerts to the farmer or veterinarian in real-time. This helps in preventing disease outbreaks, reduces treatment delays, and improves animal welfare.
IT22139894	Develop an intelligent system that predicts the likelihood of a successful and healthy birth (for calves) or successful hatching (for eggs) before the event occurs, using a combination of sensor data, images, and historical records, enabling early interventions and increasing overall farm productivity.
IT22134608	Create an intelligent livestock marketplace that helps farmers make better business decisions by predicting future prices, estimating animal value in Realtime, and recommending when to sell or buy — all supported by AI models. This enables farmers to maximize profits, reduce risks, and streamline transactions

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

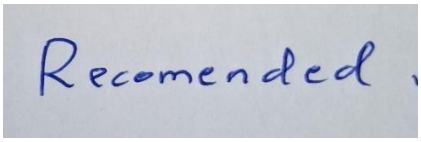
c) The student group can successfully execute the proposed project.

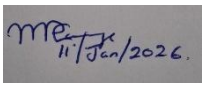
Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:



	Title	First Name	Last Name	Signature
Supervisor	Ms.	Manori	Gamage	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				